Why Did We Change the Topic?

Project ID - R26 - IT - 093

Topic	AI-Driven Early Literacy Assessment and Adaptive Reading Support System for School Children			
Project ID	R26-IT-093			
Specialization	IT			
Student ID	IT22139894	IT22136138	IT22134608	IT22282118
PanelName	IT-P07			
Detailed Comments	Need to have a proper understand about the component. Clearly highlight the novelty and individual contribution of the component by studying the existing research and discuss with supervisor.	Need to have a proper understand about the component. Clearly highlight the novelty and individual contribution of the component by studying the existing research and discuss with supervisor.	Suggestion as novelty -> Correctly pronouncing the English word in Sinhala	Need to have a proper understand about the component. Clearly highlight the novelty and individual contribution of the component by studying the existing research and discuss with supervisor.

AI SMART FARM

Project ID - R26 - IT - 093



CONTENT

1. Introdution
2. Proposed Solution User Requirements Addressed of the components
3. Design Excellence and User Feedback of the components
4. Commercialization/Sustainability of Solution



Converting Traditional farm to Smart Farm

Traditional farming still faces several major challengers,

(1) Lack of automated feeding and nutrition management in traditional livestock farming
(2) Difficulty in early detection of cattle diseases using manual observation methods
(3) Absence of reliable egg hatching and birth prediction mechanisms in conventional farms
(4) Limited access to market prediction tools for informed livestock product pricing and sales decisions

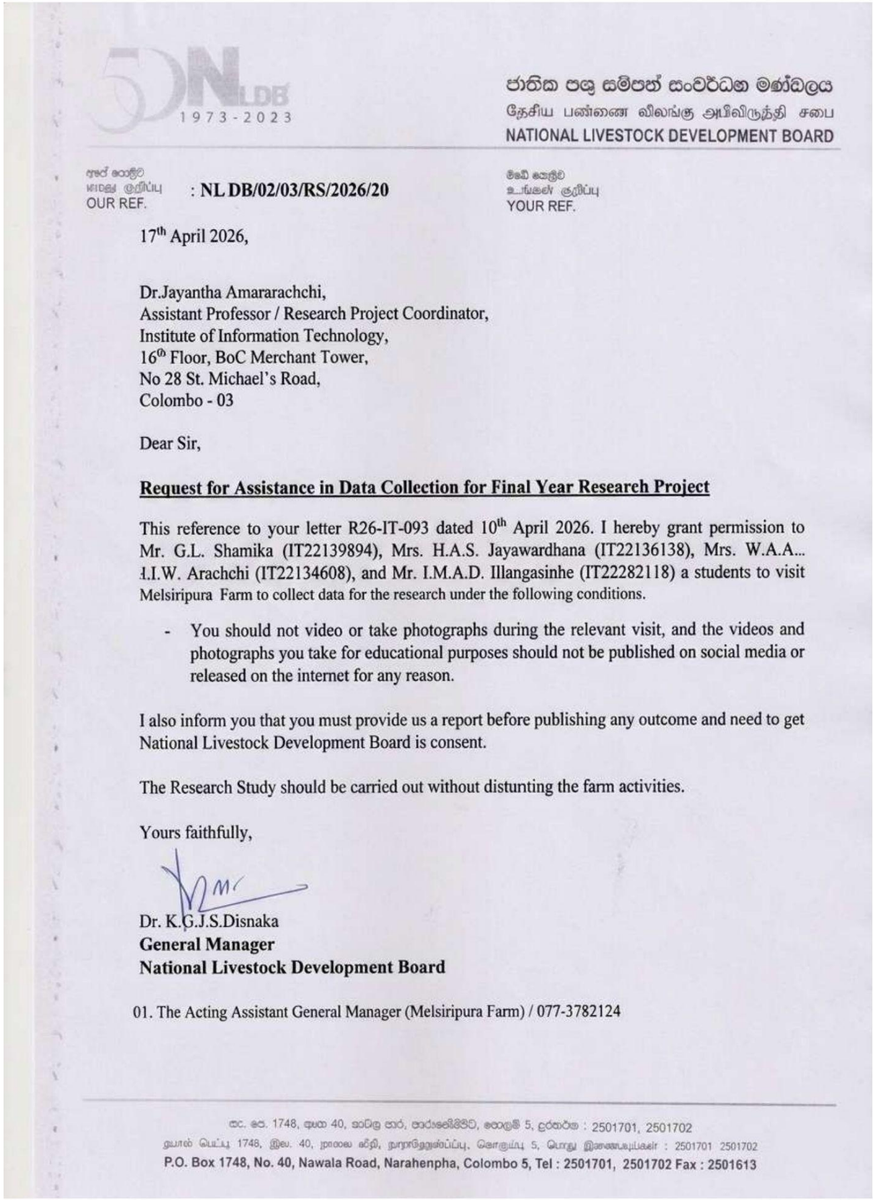
Design Excellence of the Overall Solution

- Intelligent & Automated
- Farmer-Centric
- Multi-Component Smart System
- Efficient & Time-Saving



PP1 Video

Data Collection Permission Letter



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தேசிய பண்ணை விலங்கு அபிவிருத்தி சபை
NATIONAL LIVESTOCK DEVELOPMENT BOARD

OUR REF. : NL DB/02/03/RS/2026/20

YOUR REF.

17th April 2026,

Dr.Jayantha Amararachchi,
Assistant Professor / Research Project Coordinator,
Institute of Information Technology,
16th Floor, BoC Merchant Tower,
No 28 St. Michael's Road,
Colombo - 03

Dear Sir,

Request for Assistance in Data Collection for Final Year Research Project

This reference to your letter R26-IT-093 dated 10th April 2026. I hereby grant permission to Mr. G.L. Shamika (IT22139894), Mrs. H.A.S. Jayawardhana (IT22136138), Mrs. W.A.A... I.W. Arachchi (IT22134608), and Mr. I.M.A.D. Illangasinhe (IT22282118) a students to visit Melsiripura Farm to collect data for the research under the following conditions.

- You should not video or take photographs during the relevant visit, and the videos and photographs you take for educational purposes should not be published on social media or released on the internet for any reason.

I also inform you that you must provide us a report before publishing any outcome and need to get National Livestock Development Board is consent.

The Research Study should be carried out without distunting the farm activities.

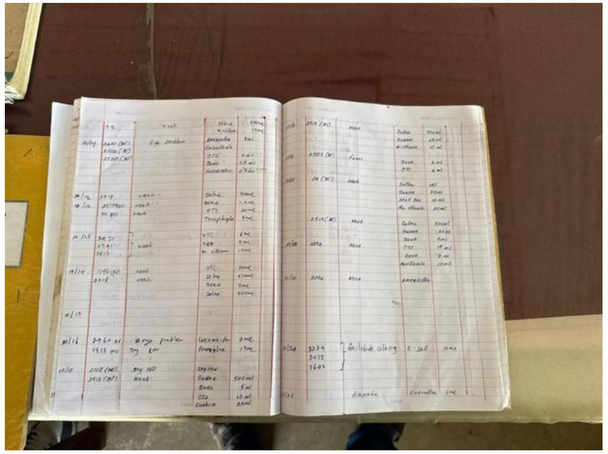
Yours faithfully,

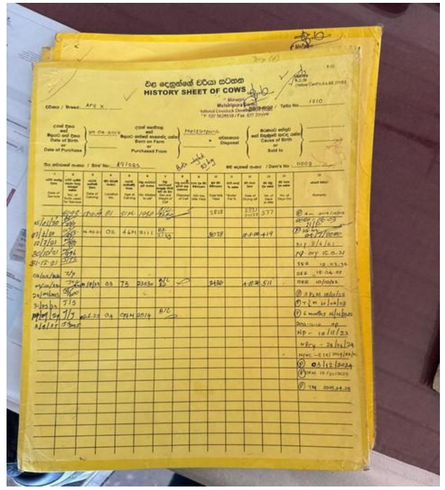
Dr. K.G.J.S.Disnaka
General Manager
National Livestock Development Board

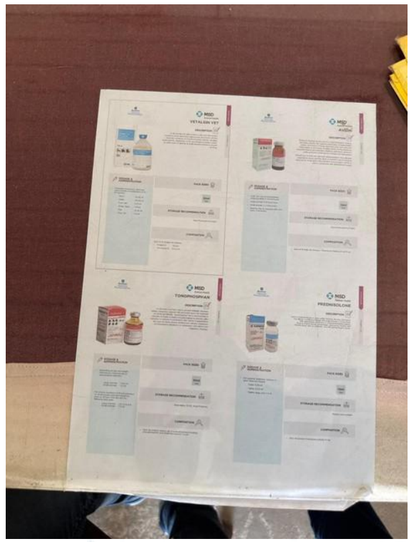
01. The Acting Assistant General Manager (Melsiripura Farm) / 077-3782124

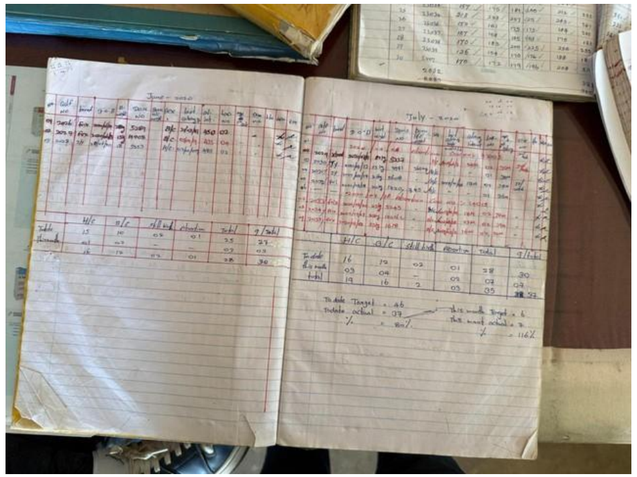
P.O. Box 1748, No. 40, Nawala Road, Narahenpha, Colombo 5, Tel : 2501701, 2501702 Fax : 2501613

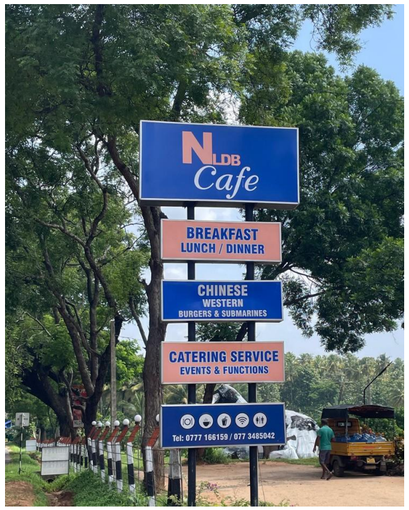
Data Collection Activities













ANIMAL BIRTH AND EGG HATCH PREDICTION

IT22139894 | Shamika G. L .| R26 – IT - 093

BSc (Hons) Degree in Information Technology specialization information Technology

Proposed Solution

- Early detection of animal birthing events
- Higher egg hatching success rate
- Increased farmer income and productivity



Modules Implemented

- Animal Birth Prediction Model
- Egg Hatch Prediction Model

DESIGN CONTRIBUTION AND USER FEEDBACK

Design Excellence

- AI-driven prediction for animal birth and egg hatching outcomes.
- Pattern-based learning from historical and real-time farm data.
- Actionable insights to support timely farming decisions.





Automated Feeding System & Nutrition Recomender

IT22136138 | Jayawardhana H.A.S. | R26 – IT - 093

BSc (Hons) Degree in Information Technology specialization information Technology

SMART FARM
EST 2026

Proposed Solution

- Maintain Cattle profile
- Receive daily feeding quantities (kg):
- Receive nutrition recommendations:

Modules Implemented

- Cattle Feeding Prediction Model
- Nutration Recomender Model

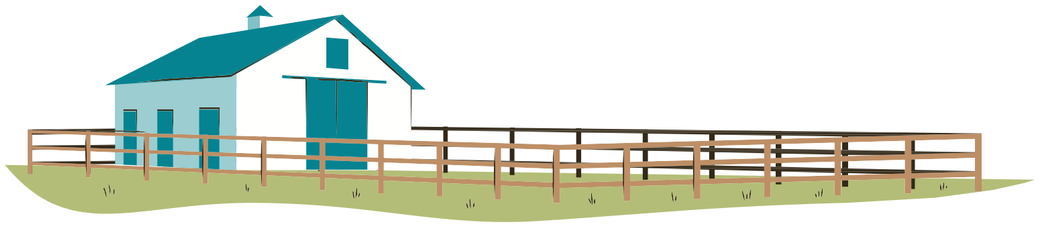


DESIGN CONTRIBUTION AND USER FEEDBACK

- Farmers were able to provide the required feed amount individually for each animal based on its weight and breed, while also reducing discomfort for the cattle.
- Farm operators valued the automated feed calculations that reduced guesswork and feed waste.

:

- Stakeholders recognized the system's potential to improve productivity through data-driven management.





Customized Live Milk Market prediction with AI/ML

IT22134608 | Arachchi W.A.A.I.W. | R26 – IT - 093

BSc (Hons) Degree in Information Technology specialization information Technology

PROPOSED SOLUTION AND USER REQUIREMENTS ADDRESSED

Proposed Solution (Milk Prediction Component):

- Multi-modal data fusion (production, pricing)
- Real-time market data collection from industry sources
- Prediction Engine: 1-month forecasting for farmers and businesses



User Requirements Addressed:

- For Farmers: Better planning & optimized product mix decisions
- For Businesses: Competitive advantage & risk management
- For Industry: Market stability & informed decision-making

DESIGN CONTRIBUTION AND USER FEEDBACK

Design Excellence

- Performance Optimization
- Community & Collaboration
- Mobile-First Design
- User-friendly & scalable

User Feedback on Prototype:

- Real-time Data Collection - Continuous performance monitoring
- Data Collection – Farm data and farmer inputs.
- Data Quality Assessment - Validation of input data sources and accuracy
- Suggested improvements- Allow predictions and reports to be accessed on low-connectivity rural networks..





Disease Detection and Emergency Alerts for Cattels

IT22282118 | Illangasinhe I.M.A.D. | R26 – IT - 093

BSc (Hons) Degree in Information Technology specialization information Technology

PROPOSED SOLUTION AND USER REQUIREMENTS ADDRESSED

Proposed Solution (Disease & Behavior Monitoring Component):

- Automated Disease Identification.
- Real-time Behavior Tracking.
- Hybrid Health Analysis.



User Requirements Addressed:

- Early Disease Detection.
- Automated Monitoring.
- Data-Driven Decisions.

DESIGN CONTRIBUTION AND USER FEEDBACK

- Farmers found the automated monitoring system easier and more convenient than traditional manual checking methods

- They also appreciated the disease detection feature because it helped reduce confusion and supported faster treatment decisions :



Converting Traditional farm to Smart Farm

Commercialization

1. Subscription-based model for small, medium, and large-scale farms
2. Tiered pricing plans based on the number of cattle and enabled features
3. Partnerships with agricultural institutions, dairy farms, and cooperatives
4. Integration with IoT devices and farm management tools



Project Progress Overview (50%+ Completion)

Completed Tasks

- Automated Daily Feeding Calculation – Feed quantity (kg/day) generated
- Disease Detection using Camera – Image-based disease identification completed
- Nutrition Recommendation Prediction – Implemented based on cattle
- parameters
- Egg Hatching & Birth Prediction – Prediction model implemented
- Milk Market Prediction System – Initial prediction model developed
- Frontend Development – ~15% completed
- Backend Development – ~50% completed
- Backend Integration – Core modules integrated with models

Project Progress Overview (50%+ Completion)

To-Do / Remaining Tasks

- Frontend Completion & UI Refinement
- (dashboards, reports, user experience improvements)
- Backend Completion
- (API optimization, validation, security)
- Market Prediction Enhancement
- (accuracy improvement and real-time data integration)
- Field Visit & Data Validation
- (on-site data collection and real-world testing)

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THANK YOU!